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Spring 2026

File System Permissions

Permissions

- Only on NTFS and ReFS volumes
- Not available on FAT or FAT32
- Permissions can be assigned to files or folders

Access Control List

- **Security Identifier (SID)** is a unique identifier for an object
- **Security Principal** is an object with a SID (user, group, computer)
- **Permission** is the right to do something (e.g., read data)
- **allow** (permission)
- **deny** (prohibition)
- **Access Control List (ACL)** is a list of security principals with assigned permissions on the object
- Each security object (file or folder) has an assigned ACL

Example:

John::List folder contents::Deny - John can't view the folder

How do permissions work?

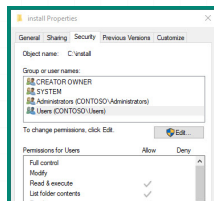
- Principle of least privilege:
- You only have explicitly assigned permissions
- No permissions mean no access (**Implicit deny**)
- Permissions of the account and the groups it belongs to are cumulative
- Prohibitions take precedence over permissions (**Explicit deny**)

Example:

Everyone::Read::Deny

Everyone::Full Control::Allow

ACL List - Basic View



ACL List - Advanced View

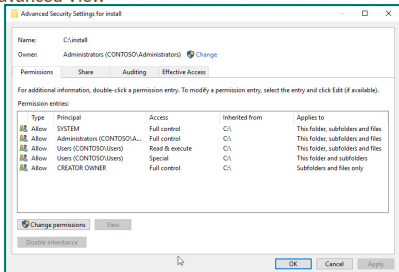


Figure 2: Properties tab > Advanced button

ACL List - Basic Permissions

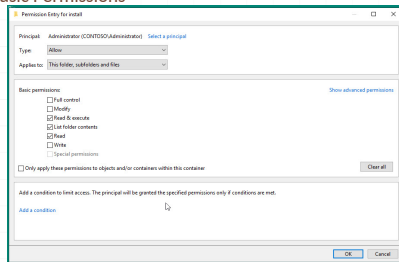
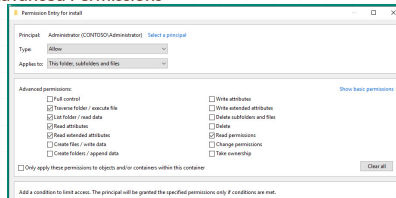


Figure 3: Basic Permissions

ACL List - Basic Permissions (2)

- Ready-made sets of advanced permissions
- Sufficient for 95-99% of everyday tasks

ACL List - Advanced Permissions

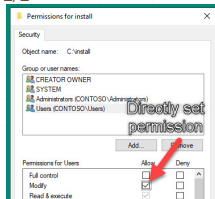


ACL List - Advanced Permissions (2)

- Some permissions can have different meanings for files and folders, e.g.:
- List folders / read data
- Used when Basic permissions are not sufficient
- Examples of use:
 - "Black hole" folder: you can upload a file, but you cannot view it
 - "Drop" folder: you can upload files, but you cannot modify them

Permission Inheritance 1/2

- Configuring multiple subfolders can be very tedious
- Inheritance – the ACL assigned to a folder is applied to subfolders and files within it
- Inherited permissions cannot be changed
 - Implemented as reference to folder with source permissions



- Inheritance can be disabled
- When disabling inheritance, previously inherited permissions can be:
 - Deleted
 - Copied and converted to direct permissions
- Tip: Do not remove permissions for the Administrators and SYSTEM accounts.
- In the Advanced permissions window, you can precisely define how each permission is inherited:

Copying and Moving Objects and Permissions

- When copying an object, it inherits permissions from the destination folder.
- If the object is moved within a volume, directly assigned permissions are retained, and inherited permissions are inherited from the destination folder.
- If the object is moved between volumes, all permissions are inherited from the destination folder.

Sharing Folders

- Accessible over the network
- Name format:

```
1 \\server_name\folder_name
```

e.g.:

```
1 \\dc1.contoso.com\profiles
```

Examples

Folder C:\reports shared as \\dc1\reports

- NTFS permissions: G_Accounting::Full control::Allow
- Share permissions: Domain Users::Read::Allow
- What are the effective permissions on the C:\reports folder?
- What are the effective permissions on the \\dc1\reports folder?

- Shared folders have two ACLs:
 - NTFS (Security tab) – discussed in the first part
 - Share (Sharing > Advanced > Permissions tab)
- NTFS permissions apply to the locally used folder (C:\profiles)
- NTFS and Share permissions apply to the remotely used folder (\\dc1\profiles)
- The lowest common part of the permissions applies

Microsoft advice

Since the introduction of the NTFS format, Microsoft advises:

- Set the Share permission to Everyone::Full control::Allow
The Share permission guarantees access for everyone, so the effective permissions are determined only by NTFS permissions, which significantly simplifies administration.

- The folder is shared under its name
- Permissions set according to Microsoft's recommendation
- All subfolders with the same permissions

See: [1] Chapter 2, section "Configure NFS and SMB shares using Server Manager"

- A folder can be shared under multiple names
- Each name (share) can have different permissions
- You can limit the number of people who can use a folder simultaneously

See: [1] Chapter 2, sections "Creating advanced shares", "Configuring share permissions", "Configure SMB share and session settings using Windows PowerShell"

- [1] C. Zacker, *Exam ref 70-740 installation, storage and compute with windows server* 2016. Microsoft Press, 2017.

